

EDA Interview Questions and Answers

1. What is the difference between Univariate, Bivariate, and Multivariate analysis?

Univariate Analysis: Analyzes one variable at a time.

Example: Histogram of age.

Bivariate Analysis: Studies the relationship between two variables.

Example: Scatter plot of height vs weight.

Multivariate Analysis: Examines relationships among multiple variables simultaneously.

Example: Pairplot or correlation matrix.

2. How do you interpret a Correlation Heatmap?

A correlation heatmap shows the strength and direction of relationships between variables.

Values close to +1 indicate a strong positive correlation.

Values close to -1 indicate a strong negative correlation.

Values near 0 indicate weak or no correlation.

3. What is the purpose of a Box plot?

A box plot is used to:

- Show data distribution
- Identify median and quartiles
- Detect outliers visually

4. How do you identify outliers visually?

Outliers can be identified using:

- Boxplots – Points outside whiskers
- Scatter plots – Distant points away from clusters
- Histograms – Extremely rare values

5. Why is EDA the most important step before modeling?

EDA helps to:

- Understand the dataset
- Detect missing values and outliers
- Identify relationships between variables
- Select useful features
- Improve model accuracy and performance

